

resistors (R1 through R5) and a variable resistor (VR1).

This circuit is based on the fact that a person's skin resistance changes when he sweats (and sweating results from lying). Dry skin has a resistance of about one million ohms, whereas the resistance of moist skin is reduced by a factor of ten or more.

Resistors R1 and R2 form a voltage divider. These have resistance values of 1 mega-ohm each and, because their values are equal, the voltage at the upper probe wire is half the battery voltage (about 4.5 volts). A person holding the probe wires will change the voltage at the upper probe wire depending on the subject's skin resistance. The skin resistance is in parallel with R2 and, because it is likely to be similar to or smaller than R2, the voltage at the probe wire will fall as skin resistance falls.

Capacitor C1 functions as a smoothing capacitor and removes the 50Hz induced mains hum that is found on a person's body. T1 and R3 form a buffer circuit (called 'emitter-follower'). The voltage at the emitter of T1 follows the voltage at the probe wire and is now able to drive transistor T2. Transistors T1 and T2 act as a voltage comparator and if the voltage at the base of T2 is higher than at the base of T3, the green LED (L1) glows. If the reverse is true, the red LED (L2) lights up.

How is the subject monitored?

The block diagram of a polygraph test indicating different sections of lie detection and the deceptive behaviour is shown in Fig. 2.

The polygraph instruments have undergone a dramatic change in the last decade. For many years, polygraphs were instruments with little needles scribbling lines on a single strip of scrolling paper. These were called 'analogue polygraphs' (Fig. 3). Today, most polygraph tests are administered with digital equipment (Fig. 4). The scrolling paper has been replaced with sophisticated algorithms



Fig. 3: An analogue polygraph instrument (photo courtesy Lafayette Instrument)



Fig. 4: Digital equipment for polygraph exams (photo courtesy Lafayette Instrument)



Fig. 5: Polygraph exam being conducted on the subject

and computer monitors.

When the subject sits down in the chair for a polygraph exam (Fig. 5), several tubes and wires are connected to his body in specific locations to monitor his physiological activities. Deceptive behaviour is supposed to trigger certain physiological changes that can be detected by a polygraph and a trained examiner, who is called a forensic psychophysicologist. This examiner looks for the extent of fluctuation in certain physiological activities.

Here's how the different physiological activities are monitored by the

polygraph.

Respiratory rate. Two pneumographs—rubber tubes filled with air—are placed around the test subject's chest and abdomen. When his chest or abdominal muscles expand, the air inside the tubes is displaced.

In an analogue polygraph, the displaced air acts on a bellows—an accordion-like device that contracts when the tubes expand. This bellows is attached to a mechanical arm, which is connected to an ink-filled pen that makes marks on the scrolling paper when the subject takes a breath.

A digital polygraph also uses the pneumographs, but employs transducers to convert the energy of the displaced air into electronic signals (Fig. 6).

Blood pressure/heart rate. A blood-pressure cuff is placed around the subject's upper arm. Tubing runs from the cuff to the polygraph. As blood pumps through the arm, it makes sound. The changes in pressure caused by the sound displace the air in the tubes. These tubes are

also connected to a bellows, which is connected to a pen that moves along with these disturbances. Again, in digital polygraphs, these signals are converted into electrical signals by transducers.

Galvanic skin resistance. Also called 'electro-dermal activity,' it is basically a measure of the sweat on the fingertips. Since the fingertips are one of the most porous areas on the body, these are a good place to look for sweat. The idea is that we sweat more when we are under stress. Fingerplates, called galvanometers, are